

I find it puzzling that although most people I talk to appreciate that good-first is better than bad-first, and many say it is obvious, they apply this gut instinct too broadly and often get the implications wrong. Let me explain with a simple thought experiment.

Assume for a moment that you have \$100,000 to invest for a few years. You place this sum in a basic mutual fund that goes on to earn 27% in the first year of ownership so that your investment is worth \$127,000 at the end of the first year. Now assume that you hold on to the fund; you don't sell any units or buy any more, and in the second year the same fund increases by a mere 7%. At the end of the second year your investment is now worth \$127,000 plus an additional 7%, which is \$135,890. Finally in the third year the fund has a very bad year and to your dismay loses 13% of its value. Your investment after three years is now worth 87% of its previous year's value, \$118,224. Out of despair and fear, you decide to get out. At least, you say to yourself, you made a total of 18.2% on your original \$100,000.

Here is my main point: What happens if I reverse the order of your investment returns and you happen to lose 13% in the first year, you earn 7% in the second year, and you get the 27% only in the third year? Will you end up with more or less than \$118,224? Are you worse off since the three years $\{-13\%, +7\%, +27\%\}$ started on the wrong foot compared to $\{+27\%, +7\%, -13\%\}$?

Many people I ask this question to say, "yes." They claim it is worse to experience the loss first. But the indisputable truth is that you will have the exact same amount of money, namely \$118,224. If you don't believe me, look at Figure 6.1 and work out the arithmetic. Notice that $\$100,000 \times 1.27 \times 1.07 \times 0.87$ is exactly the same as $\$100,000 \times 0.87 \times 1.07 \times 1.27$. The order is not important when you are buying and holding; no cash flow goes in or out. Indeed, the only thing that matters is the (compound) average of 5.7%. This is exactly why mutual funds tout their 5-, 10-, and 20-year compound returns. The year-by-year numbers don't really matter when all you do is buy and hold.

But, and many of you know this already, if you are withdrawing money from this investment, the order does become relevant, and the earlier the losses, the greater their impact. This is the so-called sequence-of-returns effect. You, like Pharaoh, want the famine returns delayed as long as possible. The objective of this chapter is to